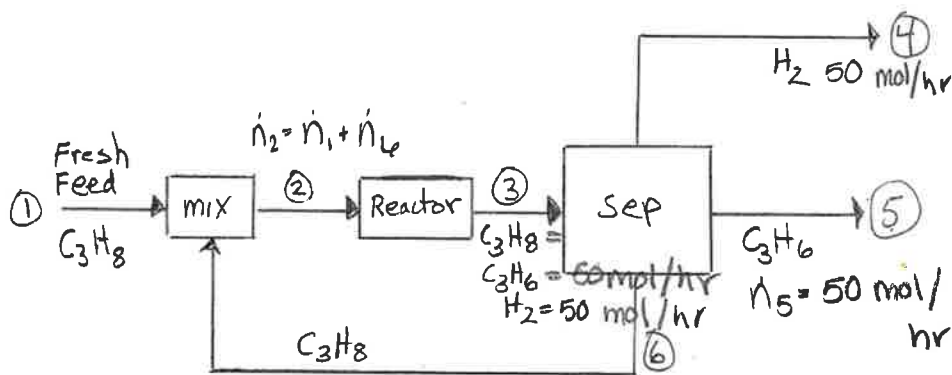
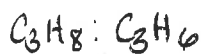


[1] (30 pts.) A process for propane dehydrogenation is shown in the figure. Assume that the only reaction taking place in the reactor is: $C_3H_8 \rightarrow C_3H_6 + H_2$ and there are no side reactions. There are six streams and the chemical species in each stream are labeled. The single-pass conversion for the reactor is 20% based on propane. The product flow rate is $\dot{n}_5 = 50 \text{ mol/hr}$ for Stream 5. Calculate all of the species molar flow rates and fill in your answers at the bottom of the page.



$$SP \ f_{con}^P = \frac{\text{moles P consumed}}{\text{mole P fed}} = 0.20$$



1:1

50 mol C_3H_6 produced = 50 mol C_3H_8 consumed

$$C_3H_6: \dot{n}_{out} = \dot{n}_{in} + \sum \dot{\xi}$$

$$\dot{n}_{out} = \sum \dot{\xi} = 50 \text{ mol/hr}$$

$$H_2: \dot{n}_{out} = \dot{n}_{in} + \sum \dot{\xi}$$

$$\dot{n}_{out} = \sum \dot{\xi} = 50 \text{ mol/hr}$$

$$SP \ f_{con}^P = \frac{50 \text{ mol } C_3H_8 \text{ consumed}}{\text{mol } C_3H_8 \text{ fed}} = 0.20$$

$$\text{mol } C_3H_8 \text{ fed} = \frac{50 \text{ mol } C_3H_8 \text{ consumed}}{0.2} = 250 \text{ mole } C_3H_8 \text{ fed from stream } \underline{2}$$

single pass is over just the reactor. Streams 2, 3

$$\dot{n}_6 = 0.8 \dot{n}_2 = (0.8)(250) = 200 \text{ mol}$$

$$\dot{n}_1 = \dot{n}_2 - \dot{n}_6 = 250 - 200 = 50 \text{ mol}$$

$$\dot{n}_{3,P} = \dot{n}_6$$

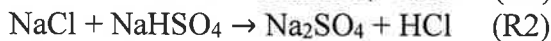
30

$$\dot{n}_1 = \underline{50} \text{ mol/hr}, \quad \dot{n}_2 = \underline{250} \text{ mol/hr}, \quad \dot{n}_{3,C_3H_8} = \underline{200} \text{ mol } C_3H_8/\text{hr},$$

$$\dot{n}_{3,C_3H_6} = \underline{50} \text{ mol } C_3H_6/\text{hr}, \quad \dot{n}_{3,H_2} = \underline{50} \text{ mol } H_2/\text{hr}, \quad \dot{n}_4 = \underline{50} \text{ mol/hr},$$

$$\dot{n}_6 = \underline{200} \text{ mol/hr}$$

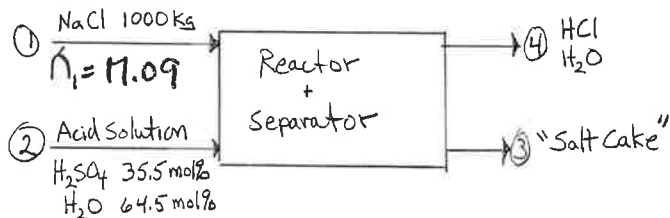
[2] (50 pts.) Salt (NaCl) and sulfuric acid (H₂SO₄) are reacted to produce hydrochloric acid (HCl) in a process represented by two chemical reactions:



For the process, there are two input streams: **Stream 1**:

1000 kg of pure NaCl and **Stream 2**: a 35.5 mol%

aqueous solution of sulfuric acid. There are two output streams: **Stream 3**: a "Salt Cake" consisting of 80.1 mol% of Na₂SO₄, 5.0 mol% of NaHSO₄, 4.2 mol% of NaCl, 9.3 mol% of H₂O, and 1.4 mol% of HCl; and **Stream 4**: a vapor stream of HCl and H₂O.



Species	Molar Mass
Na ₂ SO ₄	142.1
NaHSO ₄	120.1
NaCl	58.5
H ₂ O	18.0
HCl	36.5
H ₂ SO ₄	98.1

(a) Calculate the mole of NaCl entering the process in kmol. (5 pts.)

$$\frac{1000 \text{ kg NaCl}}{58.5 \text{ kg NaCl}} \times 1 \text{ kmol} = 17.09 \text{ kmol NaCl}$$

$$n_1 = 17.09 \text{ kmol}$$

(b) Write species mole balance equations and/or atomic balance equations. The problem can be solved by either method. (20 pts.)

$$\text{NaCl: } \dot{n}_{\text{out}} = \dot{n}_{\text{in}} - \dot{\xi}_1 - \dot{\xi}_2$$

$$0.042n_3 = 17.09 - \dot{\xi}_1 - \dot{\xi}_2$$

$$\text{H}_2\text{SO}_4: \dot{n}_{\text{out}} = \dot{n}_{\text{in}} - \dot{\xi}_1$$

$$0 = 0.355n_2 - \dot{\xi}_1$$

$$0.355n_2 = \dot{\xi}_1$$

$$\text{HCl: } \dot{n}_{\text{out}} = \dot{n}_{\text{in}} + \dot{\xi}_1 + \dot{\xi}_2$$

$$x_{\text{HCl}}n_4 + 0.014n_3 = \dot{\xi}_1 + \dot{\xi}_2$$

$$n_{4,\text{HCl}}$$

$$\text{NaHSO}_4:$$

$$\dot{n}_{\text{out}} = \dot{n}_{\text{in}} + \dot{\xi}_1 - \dot{\xi}_2$$

$$0.05n_3 = \dot{\xi}_1 - \dot{\xi}_2$$

$$\text{Na}_2\text{SO}_4:$$

$$\dot{n}_{\text{out}} = \dot{n}_{\text{in}} + \dot{\xi}_2$$

$$0.801n_3 = \dot{\xi}_2$$

$$\text{H}_2\text{O:}$$

$$\dot{n}_{\text{out}} = \dot{n}_{\text{in}}$$

$$x_{\text{H}_2\text{O}}n_4 + 0.093n_3 = 0.645n_2$$

$$n_{4,\text{H}_2\text{O}}$$

Unknowns

$$n_3$$

$$\dot{\xi}_1$$

$$\dot{\xi}_2$$

$$n_2$$

$$n_{4,\text{HCl}}$$

$$n_{4,\text{H}_2\text{O}}$$

6 equations

6 unknowns

$$\begin{array}{r|l} 1 & 30 \\ \hline 2 & 50 \\ \hline 3 & 20 \\ \hline \hline & 100 \end{array}$$

20

PROBLEM [2] CONTINUED

(c) Calculate the molar amounts of Streams 2, 3, and 4; and the mole fraction of HCl in Stream 4. (15 pts.)

$$0.05n_3 = \dot{z}_1 - \dot{z}_2 \approx 0.801n_3 = \dot{z}_2$$

$$0.05n_3 = \dot{z}_1 - 0.801n_3$$

$$0.851n_3 = \dot{z}_1$$

$$\dot{z}_1 = 0.851n_3$$

$$= 0.851(10.09)$$

$$= 8.59 \text{ kmol}$$

$$\dot{z}_2 = 0.801n_3$$

$$= 0.801(10.09)$$

$$= 8.08 \text{ kmol}$$

$$0.042n_3 = 17.09 - 0.851n_3 - 0.801n_3$$

$$1.694n_3 = 17.09$$

$$n_3 = 10.09 \text{ kmol}$$

$$0.355n_2 = \dot{z}_1$$

$$n_2 = \frac{8.59}{0.355} = 24.20 \text{ kmol}$$

$$n_{4,\text{HCl}} + 0.014(10.09) = 8.59 + 8.08$$

$$n_{4,\text{HCl}} = 16.52874$$

$$x_{4,\text{HCl}} = \frac{16.52874 \text{ kmol HCl}}{31.20 \text{ kmol tot}}$$

$$= 0.53$$

$$n_{4,\text{H}_2\text{O}} + 0.093n_3 = 0.645n_2$$

$$n_{4,\text{H}_2\text{O}} = (0.645)(24.20) - (0.093)(10.09) = 14.67063 \text{ kmol}$$

$$n_4 = n_{4,\text{HCl}} + n_{4,\text{H}_2\text{O}} = 16.52874 + 14.67063 = 31.20 \text{ kmol}$$

$$n_2 = 24.20 \text{ kmol}, n_3 = 10.09 \text{ kmol}, n_4 = 31.20 \text{ kmol}, x_{4,\text{HCl}} = 0.53$$

(d) Check the mass balance for the process. That is, show that mass in equals mass out. (10 pts.)

$$M_{\text{in}} = 1000 \text{ kg} + \left(0.355 \frac{\text{kmol H}_2\text{SO}_4}{\text{kmol tot}}\right)(24.20 \text{ kmol tot}) \left(\frac{98.1 \text{ kg H}_2\text{SO}_4}{1 \text{ kmol H}_2\text{SO}_4}\right) + \left(0.645 \frac{\text{kmol H}_2\text{O}}{\text{kmol tot}}\right)(24.20 \text{ kmol tot}) \left(\frac{18 \text{ kg H}_2\text{O}}{1 \text{ kmol H}_2\text{O}}\right)$$

$$= 1000 \text{ kg} + 842.78 + 280.96 = 2123.7391 \text{ kg}$$

$$M_{\text{salt cake}} = \left(0.801 \frac{\text{kmol Na}_2\text{SO}_4}{\text{kmol tot}}\right)(10.09 \text{ kmol tot}) \left(\frac{142.1 \text{ kg Na}_2\text{SO}_4}{1 \text{ kmol}}\right) + \left(0.05 \frac{\text{kmol NaHSO}_4}{\text{kmol tot}}\right)(10.09 \text{ kmol tot}) \left(\frac{120.1 \text{ kg}}{1 \text{ kmol}}\right) + \left(0.042 \frac{\text{kmol NaCl}}{\text{kmol tot}}\right)(10.09 \text{ kmol tot}) \left(\frac{58.5 \text{ kg NaCl}}{1 \text{ kmol}}\right) + \left(0.093 \frac{\text{kmol H}_2\text{O}}{\text{kmol tot}}\right)(10.09 \text{ kmol tot}) \left(\frac{18 \text{ kg H}_2\text{O}}{1 \text{ kmol}}\right) + \left(0.014 \frac{\text{kmol HCl}}{\text{kmol tot}}\right)(10.09 \text{ kmol tot}) \left(\frac{36.5 \text{ kg HCl}}{1 \text{ kmol}}\right) = 1255.89 \text{ kg}$$

$$M_{\text{vapor}} = (14.53 \text{ kmol HCl}) \left(\frac{36.5 \text{ kg HCl}}{1 \text{ kmol}}\right) + (14.67 \text{ kmol H}_2\text{O}) \left(\frac{18 \text{ kg H}_2\text{O}}{1 \text{ kmol}}\right) = 867.405 \text{ kg}$$

$$M_{\text{out}} = 1255.89 + 867.405 = 2123.295 \text{ kg}$$

$$M_{\text{in}} = M_{\text{out}} \text{ (w/ rounding errors) Yes!}$$

(50)

NAME Gillianne Murdock

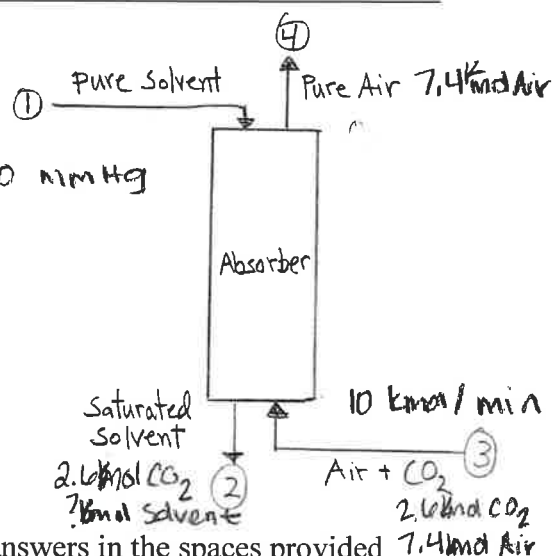
[3] (20 pts.) Carbon dioxide (CO₂) can be removed from air using an absorption process. In the figure, **Stream 1** is a pure solvent stream and **Stream 3** is an air/CO₂ mixture containing 26 mol% CO₂ at a flow rate of 10 kmol/min. The operating temperature and pressure is 50°C and 1 atm.

Assume Stream 2 and Stream 3 are in equilibrium at 50°C.

Equilibrium Data at 50°C:

Partial Pressure of CO ₂ (mmHg) in Air/CO ₂	Liquid Concentration (moles of CO ₂ /mole of solvent)
668.4	0.680
242.3	0.562
183.8	0.548
71.0	0.489
10.2	0.302

1 atm = 760 mmHg



Calculate the molar flow rates and compositions of all streams. Write your answers in the spaces provided below.

$$P_{tot} = x_{CO_2} P_{tot} + x_{Air} P_{tot}$$

$$\frac{242.3 - 183.8}{0.562 - 0.548} = \frac{197.6 - 183.8}{x - 0.548}$$

$$P_{CO_2} = x_{CO_2} P_{tot} = (0.26)(760) = 197.6 \text{ mmHg}$$

$$x - 0.548 = \left(\frac{197.6 - 183.8}{242.3 - 183.8} \right) (0.562 - 0.548)$$

$$x = \left(\frac{197.6 - 183.8}{242.3 - 183.8} \right) (0.562 - 0.548) + 0.548$$

$$\frac{n_{CO_2}}{n_{sol}} = 0.5513$$

$$\frac{2.6 \text{ kmol CO}_2}{0.5513} = n_{sol} = 4.72 \text{ kmol}$$

$$10 \text{ kmol} = 10,000 \text{ mol/min}$$

$$\dot{n}_2 = \dot{n}_{CO_2} + \dot{n}_{sol} = 2.6 + 4.72 = 7.32 \text{ kmol} = 7320 \text{ mol/min}$$

$$\dot{n}_3 = \dot{n}_{3,Air} + \dot{n}_{3,CO_2}$$

$$10 \text{ kmol} = \dot{n}_{3,Air} + (0.26)(10 \text{ kmol})$$

$$10 \text{ kmol} - 2.6 \text{ kmol} = 7.4 \text{ kmol}$$

$$\dot{n}_{3,Air} = \dot{n}_4 = 7.4 \text{ kmol} = 7400 \text{ mol/min}$$

$$\begin{aligned} \dot{n}_1 &= \dot{n}_{2,sol} \\ &= 4.72 \text{ kmol} \\ &= 4720 \text{ mol/min} \end{aligned}$$

$$x_{CO_2} = \frac{2.6 \text{ kmol CO}_2}{7.32 \text{ kmol tot}} = 0.36$$

$$\dot{n}_1 = 4720 \text{ molS/min}, \quad \dot{n}_2 = 7320 \text{ mol/min}, \quad \dot{n}_4 = 7400 \text{ molAir/min},$$

$$x_{CO_2} = 0.36 \text{ molCO}_2/\text{mol}$$

(20)